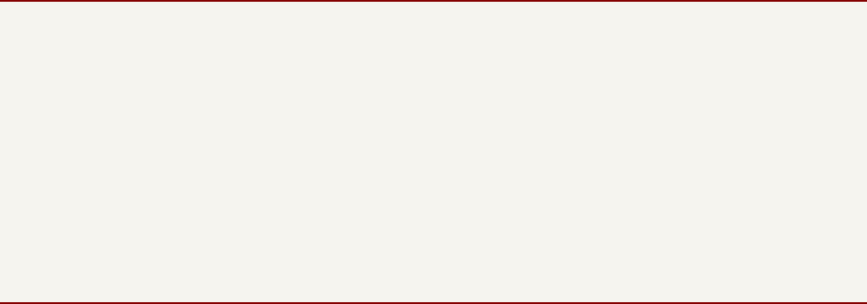


P36 | SIX-COLOR PORTRAIT BADGE

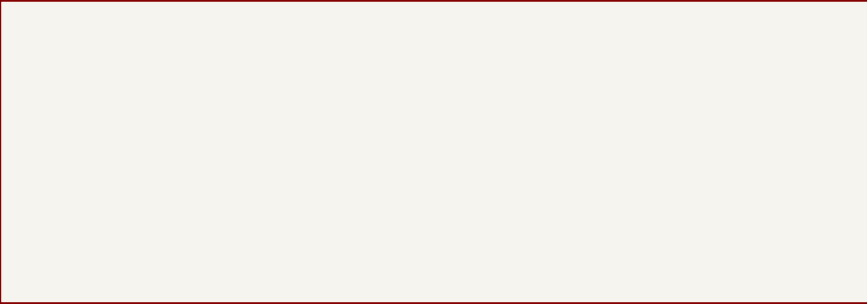
Engineering prototype – not a production release

01 USB-C + MCU



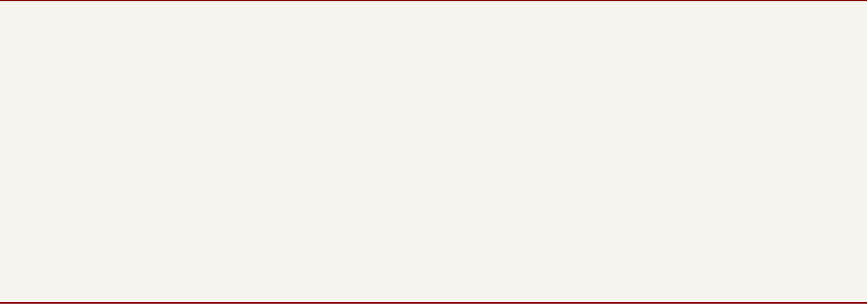
File: usb_mcu.kicad_sch

02 BATTERY + POWER



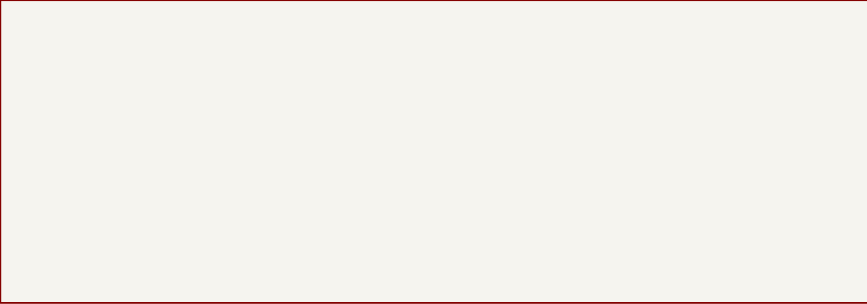
File: power.kicad_sch

03 DUAL BOOST



File: boost.kicad_sch

04 DISPLAY + FILTERS

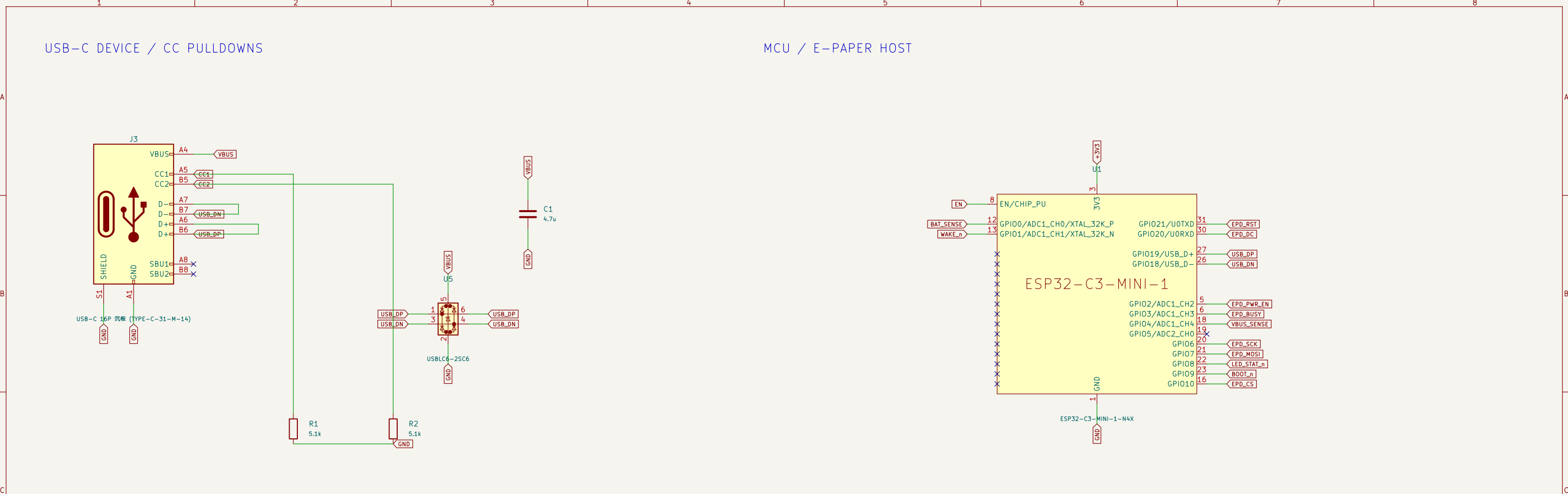


File: display.kicad_sch

Start with POWER and BOOST for electrical review.
Cross-sheet rails and MCU signals use global labels.
No external card sleeve. Internal authorised access inlay is pending system identification.
Measurements still required: refresh/USB/battery/RF/ESD/thermal/mechanical validation.



Sheet: / File: badge.kicad_sch		
Title: P36 – Engineering review hierarchy		
Size: A3	Date: 2026-09-30	Rev: P36 prototype
KiCad E.D.A. 10.0.6	Id: 1/5	

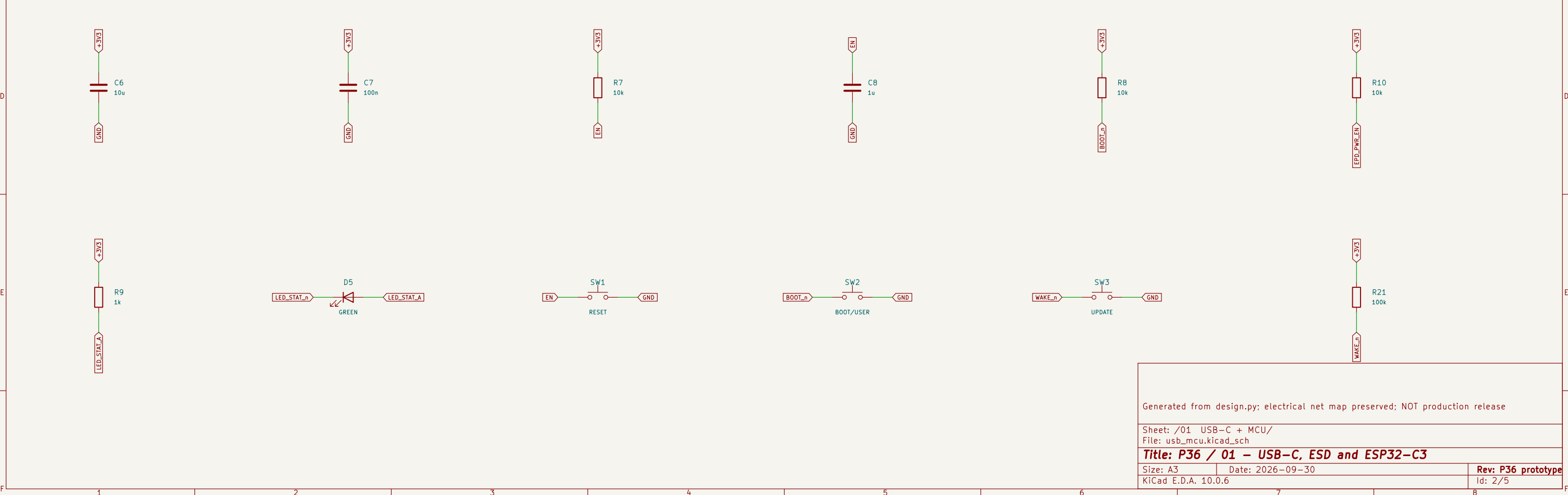


Generated from design.py; electrical net map preserved; NOT production release

Sheet: /01 USB-C + MCU/
File: usb_mcu.kicad_sch

Title: P36 / 01 - USB-C, ESD and ESP32-C3

Size: A3	Date: 2026-09-30	Rev: P36 prototype
KiCad E.D.A. 10.0.6		Id: 2/5

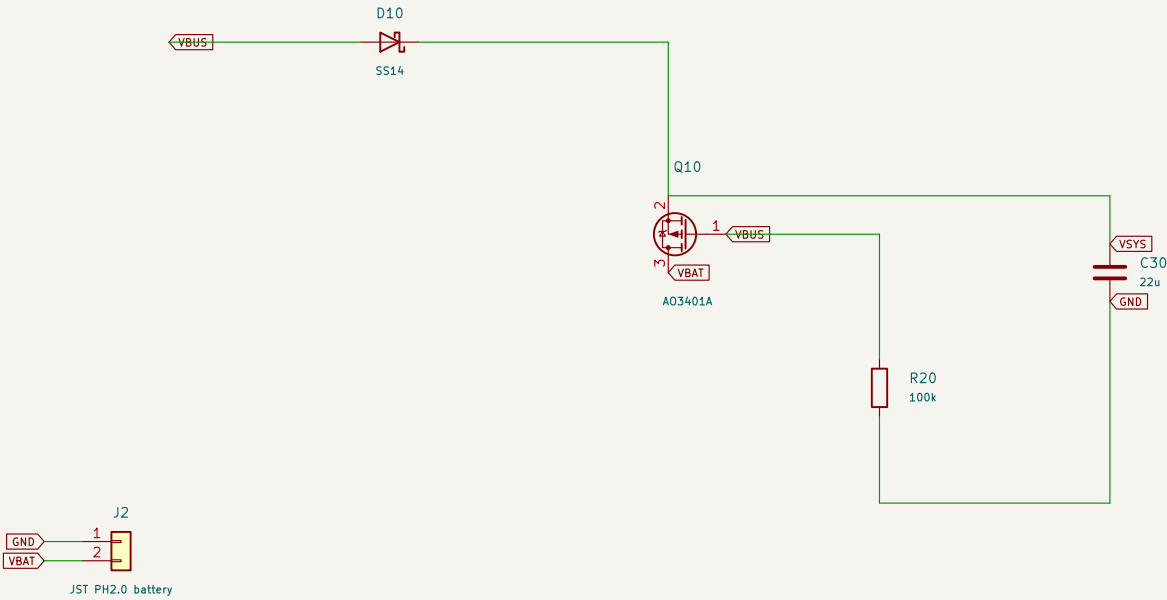


Sheet: /01 USB-C + MCU/ File: usb_mcu.kicad_sch		
Title: P36 / 01 – USB-C, ESD and ESP32-C3		
Size: A3	Date: 2026-09-30	Rev: P36 prototype
KiCad E.D.A. 10.0.6	Id: 2/5	

Sheet: /01 USB-C + MCU/ File: usb_mcu.kicad_sch		
Title: P36 / 01 – USB-C, ESD and ESP32-C3		
Size: A3	Date: 2026-09-30	Rev: P36 prototype
KiCad E.D.A. 10.0.6	Id: 2/5	

USB / BATTERY LOAD SHARING

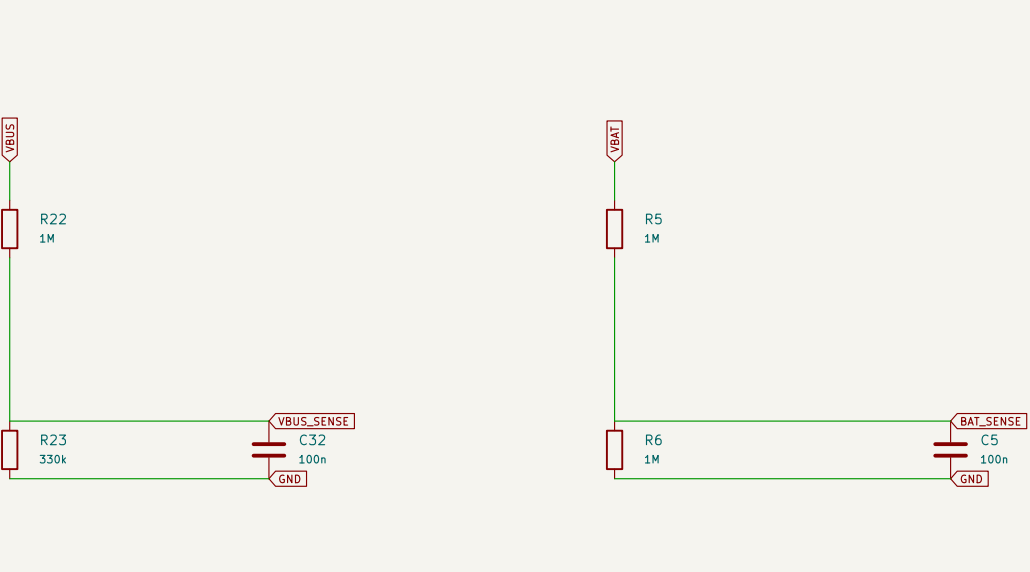
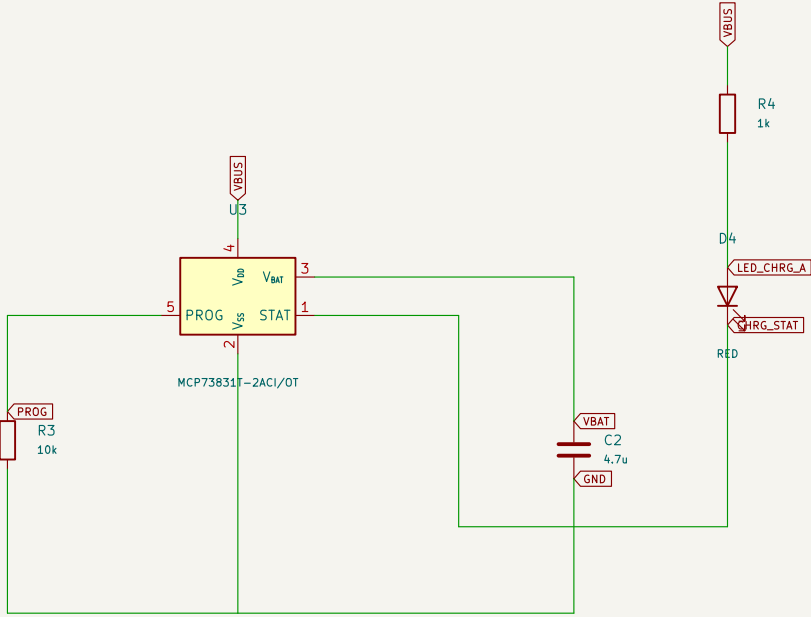
3.3 V REGULATOR



No cell-temperature input; prototype charging only in controlled 0–45 C conditions.
Charger thermal regulation protects its die, not the battery. Verify enclosure temperature.
At low VBAT, LDO dropout limits usable capacity; test simultaneous RF + refresh load.

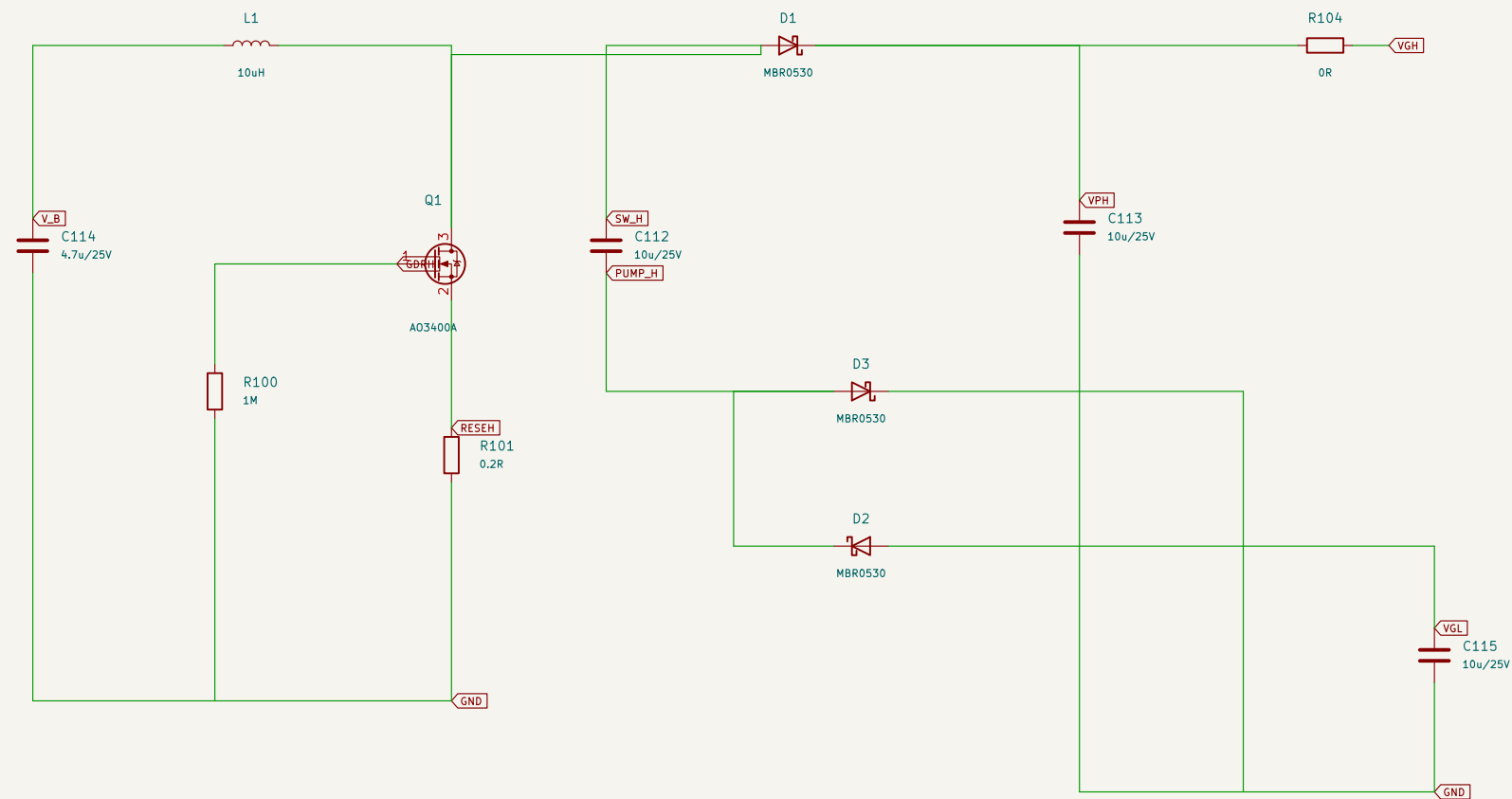
MCP73831 CHARGER / 100 mA NOMINAL

BATTERY / USB PRESENT ADC



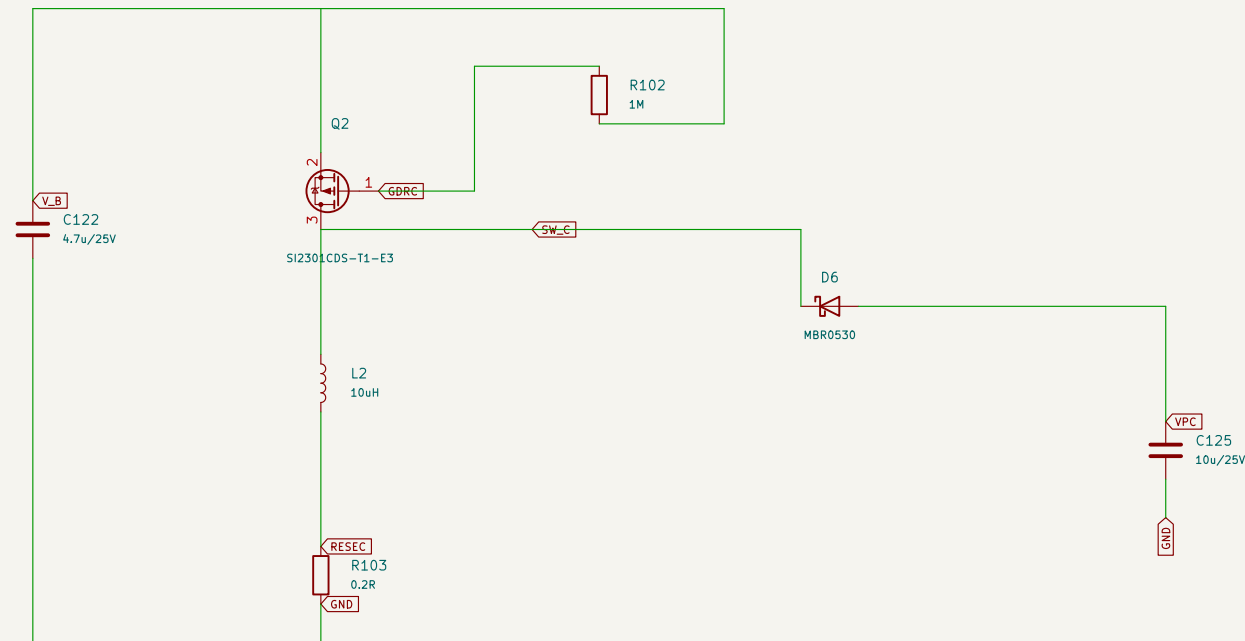
Generated from design.py; electrical net map preserved; NOT production release

POSITIVE BOOST + INVERTING CHARGE PUMP



Reference: Waveshare 3.6 E6 HAT+ schematic, Preserve 10 uH / 0.2 ohm values.
Scope boost rails, startup ripple, MOS drive and effective MLCC capacitance before qualification.
V_B bulk reservoir and the separated logic ferrite branch are on the display sheet.

NEGATIVE BOOST / P-MOS SOURCE TOWARD V_B



Generated from design.py; electrical net map preserved; NOT production release

Sheet: /03 DUAL BOOST/ File: boost.kicad_sch		
Title: P36 / 03 – E6 positive and negative boost reference		
Size: A3	Date: 2026-09-30	Rev: P36 prototype
KiCad E.D.A. 10.0.6		Id: 4/5

3.6 INCH SPECTRA 6 / 400 x 600 PORTRAIT

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TFT_VCOM
FPL_VCOM
GDRH
RESEH
GND
GDRH
RESEC
VPC
GND
VGL
VPH
VSH
VSH_LV
VSH_LV2
VSL
VSL_LV
VSL_LV2
GND
REFN
REFP
GND
GND
EPD_RST
EPD_BUSY
EPD_DC
EPD_CS
EPD_SCK
EPD_MOSI
VDDDD
EPD_3V3
GND
EPD_3V3
VCP2
CP2N
CP2P
VCP1
CP1N
CP1P
VGH
VCOMBD

EPD50 FH34SRJ

Q20
A03401A
EPD_PWR_EN
EPD_SUPPLY
FB2
BLM18PG1215N10
FB1
BLM18PG1215N10
EPD_3V3
C101
100n
V_B
C100
100u/6.3V
GND

C110
4.7u/25V
C111
470n/25V
C116
10u/25V
C117
10u/25V
C118
10u/25V
C119
10u/25V
C120
10u/25V
C121
10u/25V
C123
2.2u/6.3V
C124
1u/50V
C126
1u/50V
C127
1u/50V
C128
1u/50V
C129
470n/25V
C130
1u/50V
C131
1u/50V
GND

TFT_VCOM
FPL_VCOM
VSH
VSH_LV
VSL
VSL_LV
VDDDD
VCP2
CP2N
CP2P
VCP1
CP1N
CP1P
VCOMBD
REFN
REFP
GND

FH34SRJ-50S-0.5SH(50)
50 pin / 0.5 mm / 0.3 mm FPC
Pin 1 and contact orientation: inspect real panel
BS1 = 0, BS0 = 0: four-wire SPI
Only specified NC pins left open

All boost capacitors retain vendor voltage ratings.
MLCC nominal capacitance is not its capacitance under DC bias.
No universal door-access electronics: authorised internal credential remains system-dependent.

Generated from design.py; electrical net map preserved; NOT production release

Sheet: /04 DISPLAY + FILTERS/
File: display.kicad_sch

Title: P36 / 04 - E6 50-pin interface and rail filtering

Size: A3 Date: 2026-09-30 Rev: P36 prototype
KiCad E.D.A. 10.0.6 Id: 5/5

3.6 INCH SPECTRA 6 / 400 x 600 PORTRAIT

J1

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TFT_VCOM
FPL_VCOM
GDRH
RESEH
GND
GDRH
RESEC
VPC
GND
VGL
VPH
VSH
VSH_LV
VSH_LV2
VSL
VSL_LV
VSL_LV2
GND
REFN
REFP
GND
GND
EPD_RST
EPD_BUSY
EPD_DC
EPD_CS
EPD_SCK
EPD_MOSI
VDDDD
EPD_3V3
GND
EPD_3V3
VCP2
CP2N
CP2P
VCP1
CP1N
CP1P
VGH
VCOMBD

EPD50 FH34SRJ

Q20
A03401A
EPD_PWR_EN
EPD_SUPPLY
FB2
BLM18PG1215N10
FB1
BLM18PG1215N10
EPD_3V3
C101
100n
V_B
C100
100u/6.3V
GND

C110
4.7u/25V
C111
470n/25V
C116
10u/25V
C117
10u/25V
C118
10u/25V
C119
10u/25V
C120
10u/25V
C121
10u/25V
C123
2.2u/6.3V
C124
1u/50V
C126
1u/50V
C127
1u/50V
C128
1u/50V
C129
470n/25V
C130
1u/50V
C131
1u/50V
GND

TFT_VCOM
FPL_VCOM
VSH
VSH_LV
VSL
VSL_LV
VDDDD
VCP2
CP2N
CP2P
VCP1
CP1N
CP1P
VCOMBD
REFN
REFP
GND

FH34SRJ-50S-0.5SH(50)
50 pin / 0.5 mm / 0.3 mm FPC
Pin 1 and contact orientation: inspect real panel
BS1 = 0, BS0 = 0: four-wire SPI
Only specified NC pins left open

All boost capacitors retain vendor voltage ratings.
MLCC nominal capacitance is not its capacitance under DC bias.
No universal door-access electronics: authorised internal credential remains system-dependent.

Generated from design.py; electrical net map preserved; NOT production release

Sheet: /04 DISPLAY + FILTERS/
File: display.kicad_sch

Title: P36 / 04 - E6 50-pin interface and rail filtering

Size: A3 Date: 2026-09-30 Rev: P36 prototype
KiCad E.D.A. 10.0.6 Id: 5/5

3.6 INCH SPECTRA 6 / 400 x 600 PORTRAIT

J1

TFT_VCOM
FPL_VCOM
GDRH
RESEH
GND
GDRC
RESEC
VPC
GND
VGL
VPH
VSH
VSH_LV
VSH_LV2
VSL
VSL_LV
VSL_LV2
GND
REFN
REFP
GND
EPD_RST
EPD_BUSY
EPD_DC
EPD_CS
EPD_SCK
EPD_MOSI
GND
VDDDO
EPD_3V3
GND
EPD_3V3
VCP2
CP2N
CP2P
VCP1
CP1N
CP1P
GND
VGHI
VCOMBD

EPD50 FH34SRJ

FH34SRJ-50S-0.5SH(50)
50 pin / 0.5 mm / 0.3 mm FPC
Pin 1 and contact orientation: inspect real panel
BS1 = 0, BS0 = 0: four-wire SPI
Only specified NC pins left open

All boost capacitors retain vendor voltage ratings.
MLCC nominal capacitance is not its capacitance under DC bias.
No universal door-access electronics: authorised internal credential remains system-dependent.

Q20
AO3401A
+3V3
EPD_PWR_EN
FB2
BLM18PG121SN10
EPD_SUPPLY
FB1
BLM18PG121SN10
V_B
C100
100u/6.3V
EPD_3V3
C101
100n
GND

C110
4.7u/25V
TFT_VCOM
C111
470n/25V
FPL_VCOM
C116
10u/25V
VSH
C117
10u/25V
VSH_LV
C118
10u/25V
VSH_LV2
GND

C119
10u/25V
VSL
C120
10u/25V
VSL_LV
C121
10u/25V
VSL_LV2
C123
2.2u/6.3V
VDDDO
C124
1u/50V
VCP2
GND

C126
1u/50V
CP2P
C127
1u/50V
VCP1
C128
1u/50V
CP1P
C129
470n/25V
VCOMBD
C130
1u/50V
REFN
GND

C131
1u/50V
REFP
GND

Generated from design.py; electrical net map preserved; NOT production release

Sheet: /04 DISPLAY + FILTERS/
File: display.kicad_sch

Title: P36 / 04 - E6 50-pin interface and rail filtering

Size: A3 Date: 2026-09-30 Rev: P36 prototype
KiCad E.D.A. 10.0.6 Id: 5/5

Sheet: /04 DISPLAY + FILTERS/		
File: display.kicad_sch		
Title: P36 / 04 – E6 50-pin interface and rail filtering		
Size: A3	Date: 2026-09-30	Rev: P36 prototype
KiCad E.D.A. 10.0.6		Id: 5/5